ASEC Android Updates

How to send a fix with Shorebird, publish a new version, and force every user to update.

The two kinds of update

	Shorebird patch	New Google Play version
Use it for	Fixes and changes in Dart code, texts, images/assets	Native Android changes, new or upgraded packages, version number change
Command	shorebird patch android	shorebird release android + upload .aab to Play
Play review	No	Yes
What users see	Downloads in the background, then an **"Update ready"** dialog. Applied the next time the app opens.	**"New version available"** dialog (or the forced **"Update required"** dialog) that opens Google Play.

> **Every time**
>
> Before any command, open the terminal in the project and make sure Shorebird is on the PATH (only needed if you didn't reopen the terminal after installing):

```
cd ~/Desktop/ASEC
export PATH="$HOME/.shorebird/bin:$PATH"
shorebird --version
```

Part A - Send a fix with Shorebird (no Play upload)

1. Make the change in Dart code and test it

```
flutter run
```

2. Send the patch

Patches the latest release (the version that is on Play now):

```
shorebird patch android
```

Or choose the exact version to patch (the **version:** line in pubspec.yaml of that release):

```
shorebird patch android --release-version=1.0.1+5
```

3. What happens on the phones

Next time a user opens the app, the patch downloads in the background and the **"Update ready"** dialog appears with **Close & reopen** / **Later**. The fix is active the next time the app is opened.

4. Something went wrong? Roll back

Open **console.shorebird.dev** -> app **asec** -> the release -> the patch -> **Rollback**. Users return to the previous code on their next launch.

> **Rules**
>
> A patch can NOT contain: changes in the android/ folder, AndroidManifest, new or updated packages in pubspec.yaml, or a different version number. Shorebird stops and tells you if it finds native changes - then make a new Play version (Part B).
>
> Patches only reach phones that installed a version built with **shorebird release** (1.0.1+5 and later). Versions built with flutter build never get patches.

Part B - Publish a new Google Play version

1. Raise the version in pubspec.yaml

The number after **+** is the **build number**. It must go up every time.

```
version: 1.0.1+5      # now
version: 1.0.2+6      # next release
```

2. Build with Shorebird (not flutter build)

```
shorebird release android
```

3. Upload to Google Play Console

Upload **build/app/outputs/bundle/release/app-release.aab** -> create the release -> roll it out to 100%. Wait until Play shows it as live (can take from minutes to a few hours for review).

Once live, users on older versions see the optional **"New version available"** dialog automatically (Google Play reports the update). To force it, continue with Part C.

Part C - Show the update dialog to ALL users / force the update

The app reads one Firestore document on every launch: **settings / app_update**. You control the dialog from the Firebase console - no new build needed.

1. Publish the new version on Play first (Part B) and wait until it is live

Otherwise users are sent to Play and find nothing to update.

2. Open the document in Firebase

console.firebase.google.com -> your project -> **Firestore Database** -> collection **settings** -> document **app_update**. If it doesn't exist: **+ Add document**, Document ID = **app_update**.

3. Set the fields

Field	Type	What it does
min_build	number	**FORCED.** Every phone with a build number LOWER than this sees "Update required". The dialog has no Later button and can't be closed. Only "Update now".
latest_build	number	**OPTIONAL dialog.** Phones below this see "New version available" with Update now / Later.
notes_en	string	"What's new" text shown in the dialog (English).
notes_ar	string	"What's new" text shown in the dialog (Arabic).

Example - you just published **1.0.2+6** and everyone must move to it:

Field	Type	Value
min_build	number	6
notes_en	string	Faster videos and quiz fixes.
notes_ar	string	(your Arabic text)

Result: anyone on build 5 or lower gets the forced dialog the next time they open the app. **Update now** opens Google Play's update screen inside the app (or the ASEC page on Play). If they back out of Play, the dialog is still there.

4. Optional instead of forced

Use **latest_build = 6** and leave **min_build** out. Users can tap **Later**.

5. Turn it off

Set **min_build** to **0** or delete the field. Takes effect on the next app launch.

Important

Number type, not string. In Firebase choose type **number** for min_build / latest_build. A string "6" is ignored.

Never set min_build above the version that is on Play. Users would be stuck on a dialog with nothing to install.

Only builds 1.0.1+5 and later have this dialog. Phones still on 1.0.0+4 or older don't contain the code, so they can't be forced - they only get Play's normal auto-update.

Testing: use the **Internal testing** track on Play. The Play part of the check only works when the app was installed from Play (not from flutter run or a copied APK). The Firestore min_build part works on any release build.

Cheat sheet

I want to...	Do this
Fix a bug in Dart code	`shorebird patch android`
Patch a specific version	`shorebird patch android --release-version=1.0.1+5`
Try a release on my phone	`shorebird preview`
Undo a bad patch	console.shorebird.dev -> release -> patch -> Rollback
Ship a new Play version	Raise version in pubspec -> `shorebird release android` -> upload .aab
Ask users to update (optional)	Firestore settings/app_update: latest_build = new build number
Force everyone to update	Firestore settings/app_update: min_build = new build number
Stop forcing	min_build = 0 (or delete the field)